

Laboratory session 1

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1 Laboratory session 1

The first experiment is using standing waves inside a long tube in order to measure the speed of sound inside the gas the tube is filled with. The tube and experiment settings are specified by the following data:

Tube length: $L = (1244 \pm 1.5)mm$.

input signal: $V_0 = 35mV$

Tubes:

K1: argon or CO_2 at $T = T_{room}$

K2: air at $T = T_{room}$

K3: air at $T = 70^\circ C = 343.15K$

K4: air at $T = 50^\circ C = 323.15K$

$T_{room} = 21^\circ C = 294.15K$

We analyzed the properties of the gas in tube K4. The resistance of the tube is measured to be $R = (34.55 \pm 0.01)k\Omega$.

2 Tasks

2.1 Measurement of ν_k

We measured the following resonance frequencies ν_k between $200Hz$ and $2kHz$:

K4: $\nu_k = [313 \pm 1, 450 \pm 1, 594 \pm 1, 740 \pm 1, 886 \pm 1, 1033 \pm 1, 1178 \pm 1, 1326 \pm 1, 1473 \pm 1, 1620 \pm 1, 1767 \pm 1, 1913 \pm 1]$

K3: $\nu_k = [301 \pm 1, 459 \pm 1, 607 \pm 1, 757 \pm 1, 907 \pm 1, 1057 \pm 1, 1208 \pm 1, 1359 \pm 1, 1510 \pm 1, 1661 \pm 1, 1813 \pm 1, 1963 \pm 1]$

The other groups measured:

K2: $\nu_k = [286 \pm 1, 419 \pm 1, 470 \pm 1, 563 \pm 1, 700 \pm 1, 840 \pm 1, 979 \pm 1, 1117 \pm 1, 1258 \pm 1, 1397 \pm 1, 1537 \pm 1, 1677 \pm 1, 1816 \pm 1, 1955 \pm 1]$

K1: argon: $\nu_k = [265 \pm 1, 393 \pm 1, 521 \pm 1, 650 \pm 1, 778 \pm 1, 905 \pm 1, 1035 \pm 1, 1164 \pm 1, 1293 \pm 1, 1423 \pm 1, 1551 \pm 1, 1680 \pm 1, 1810 \pm 1, 1938 \pm 1]$

CO_2 : $\nu_k = [225 \pm 1, 440 \pm 1, 544 \pm 1, 656 \pm 1, 764 \pm 1, 872 \pm 1, 979 \pm 1, 1091 \pm 1, 1199 \pm 1, 1209 \pm 1, 1418 \pm 1, 1527 \pm 1, 1635 \pm 1, 1745 \pm 1, 1854 \pm 1, 1963 \pm 1]$

2.2 Estimated error

We estimated our error to be around $1Hz$.

2.3 The best linear fit

In order to find the speed of sound in the gas, we want to use the equation $\nu_n = an + b$. However, we do not know which resonance frequency we are measuring, so we do not know which n belongs to which measured ν_n . This is why we set the reference of the measured ν_n to the first one measured. Therefore we get:

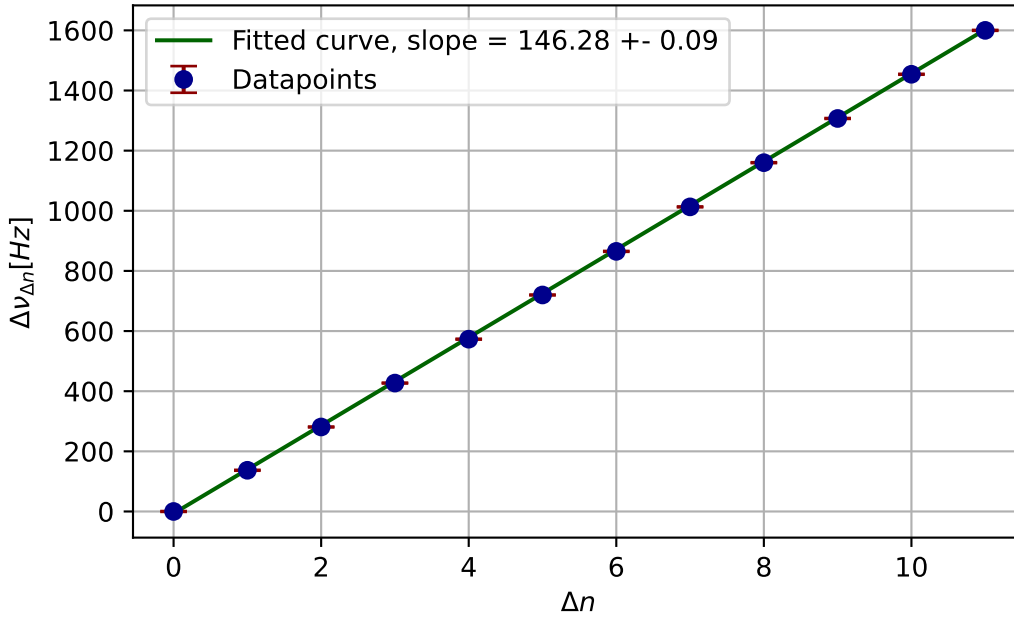
$$\nu_n - \nu_{n_0} = \Delta\nu_{\Delta n} = a\Delta n + b - (a(n_0) + b) = a(n - n_0) + b - b = a\Delta n = \frac{c}{2L}\Delta n$$

So the central relation is:

$$\Delta\nu_{\Delta n} = \frac{c}{2L}\Delta n$$

where Δn starts at 0 for the first measurement and for k measurements continues until $k - 1$ for the k^{th} resonance frequency. $\Delta\nu_{\Delta n}$ against Δn is plotted and evaluated in the following:

K4:

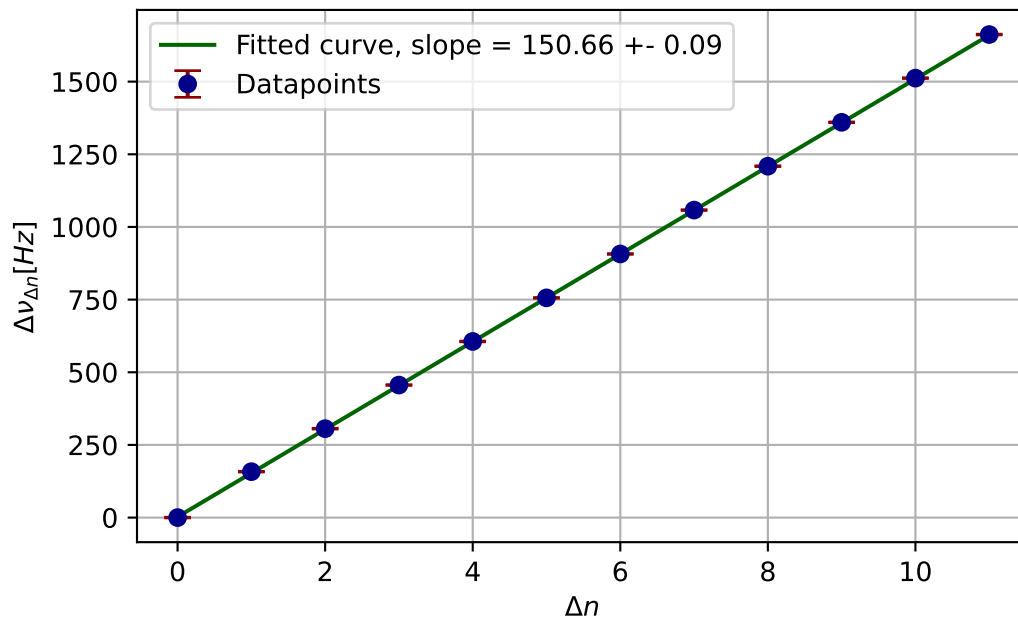


Given the slope of the fit curve, the velocity of sound can be calculated:

$$\text{speed of sound} = 363.95 \pm 0.49 \text{ m/s}$$

K3:

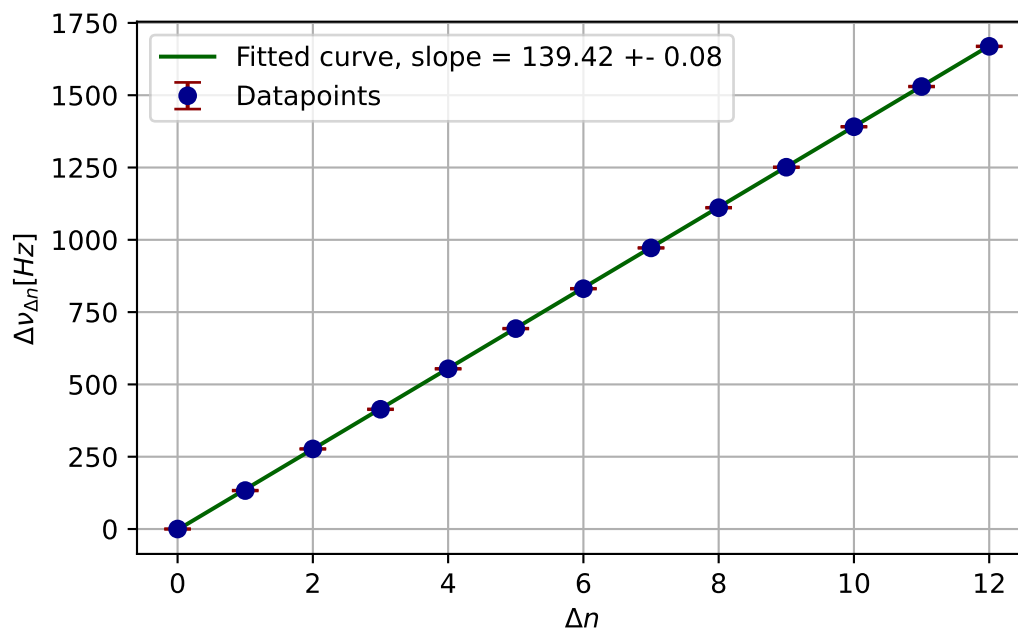
$$\text{The resistance of the tube is measured to be } R = (15.66 \pm 0.02)k\Omega$$



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 374.85 ± 0.50 m/s

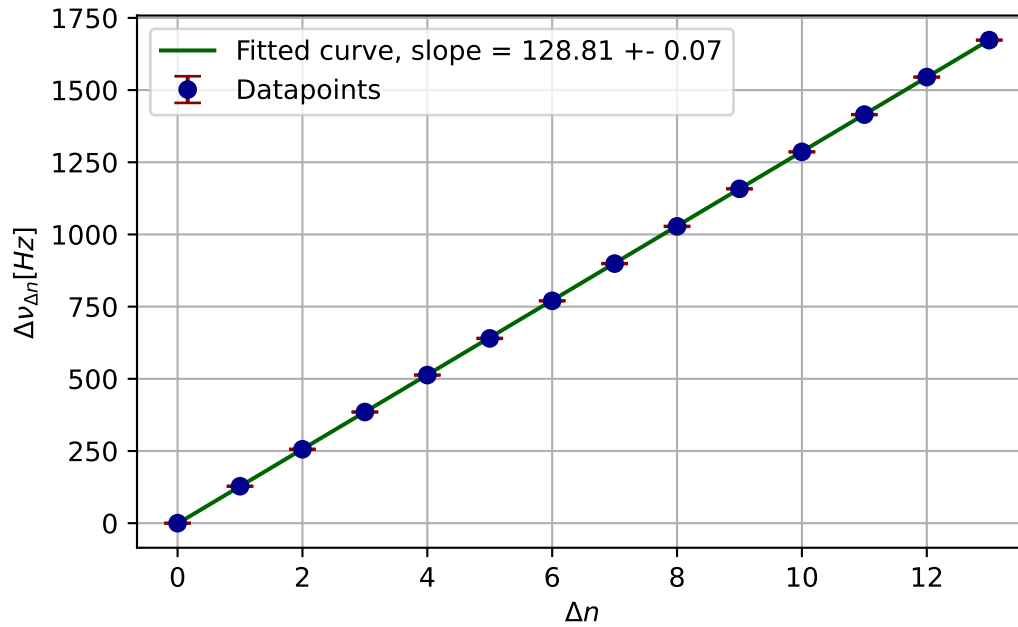
K2:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 346.87 ± 0.46 m/s

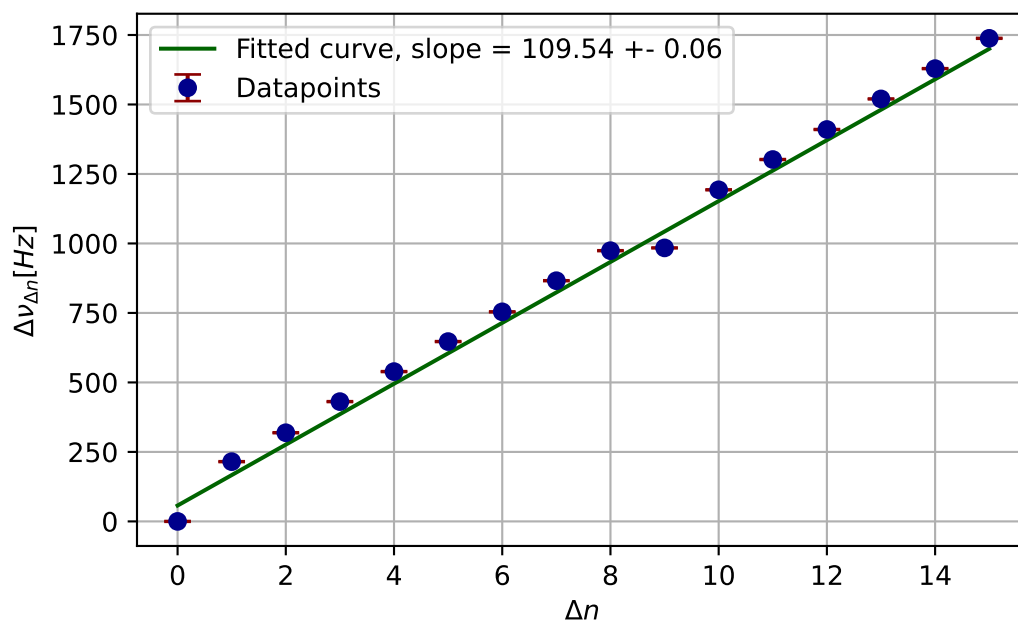
K1, argon:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 320.47 ± 0.42 m/s

K1, CO_2 :



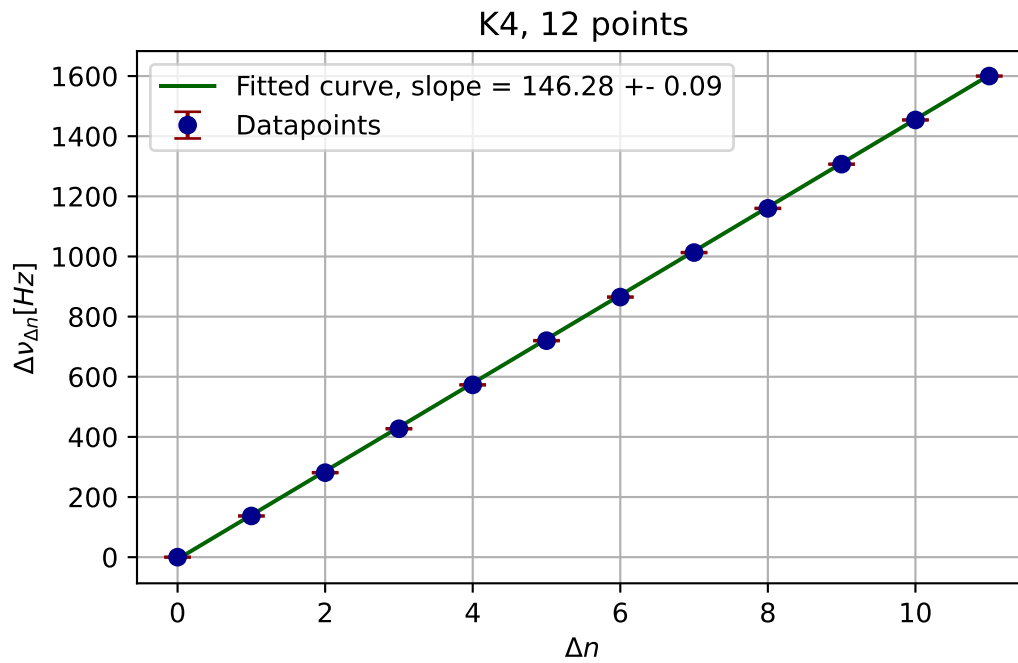
Given the slope of the fit curve, the velocity of sound can be calculated:

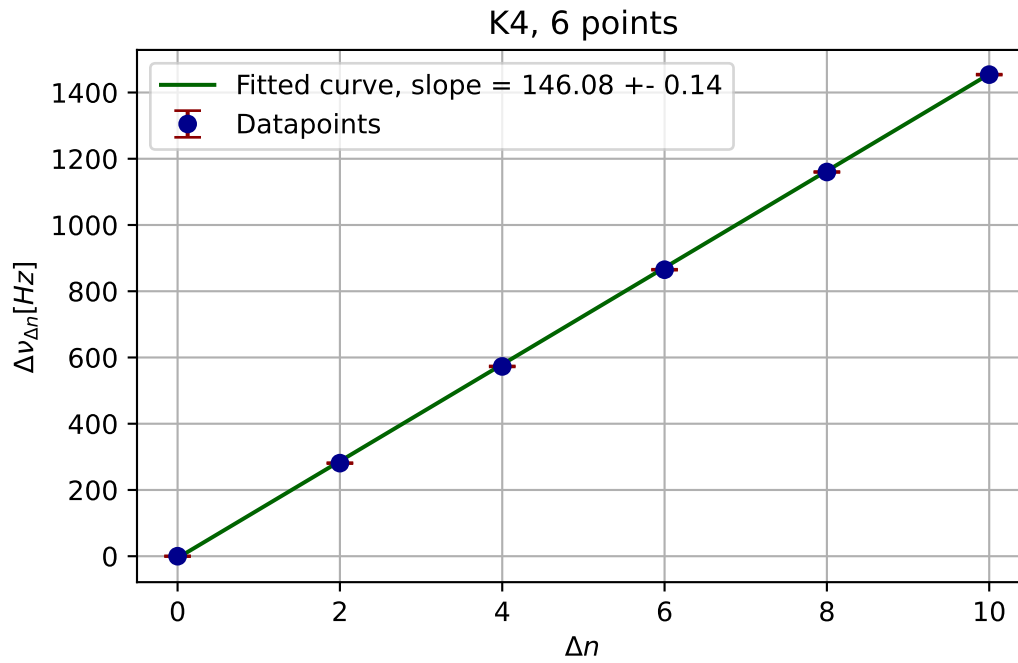
speed of sound = 272.53 ± 0.36 m/s

2.4 Most important contribution to uncertainty

The most important contribution to the error is the inaccuracy in finetuning the resonance frequency. We estimated our accuracy in doing so to $1Hz$.

The uncertainty shrinks the more points are included in the fit. In the following we see the plot for K4, once with 12 points and once with 6:





As we can see, the uncertainty on the slope is more than 50% greater if we only choose half of the points.

2.5 Degrees of freedom

For an ideal gas, the velocity of sound is given by

$$c(T) = \sqrt{\frac{(f + 2)RT}{fM_{mol}}}$$

where f are the degrees of freedom, R is the gas constant, T the temperature of the gas and M_{mol} the molar mass. This equation can be solved for f :

$$f = \frac{2RT}{c^2 M_{mol} - RT}$$

We calculate for f :

K4:

The calculated amount of degrees of freedom from the experiment is 5.81 ± 0.09

K3:

The calculated amount of degrees of freedom from the experiment is 3.89 ± 0.05

K2:

The calculated amount of degrees of freedom from the experiment is 4.71 ± 0.07

K1, argon:

The calculated amount of degrees of freedom from the experiment is 2.94 ± 0.03

K1, co2:

The calculated amount of degrees of freedom from the experiment is 5.95 ± 0.10

In theory, the air inside K2, K3, K4 should have 5 degrees of freedom. These consist of 3 for translation and 2 for rotation. CO2 can also be described as a 2-atom molecule because it is linear, therefore it should also have 5 degrees of freedom. Argon is a one-atom gas and only has 3 translational degrees of freedom.

However, our measurement for the degrees of freedom does not fit the theoretical expectation. This might be because the measurement of the speed of sound we used to calculate this is not very accurate. We fixed the measurement for K2 a bit by not using one value that was obviously off and improving the linear regression.

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3 Simulation

3.1 Phase diagram

We change the parameters in the simulation and observe how the simulation changes. Therefore, we double the initial values and plot the simulation's output.

As seen in the plots, the pressure is proportional to the starting temperature, which makes sense considering the ideal gas law. Furthermore, the compressibility factor Z is between 0.7 and 1.1 which is quite close to one. This means that we can assume our gas to be rather ideal.

The density $\rho^* = 0.1$ might have been a bit too high and therefore the Lennard-Jones gas does not act completely as an ideal gas.

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3.2 C_V , Z and degrees of freedom of a simulated Lennard-Jones gas

We used a Lennard-Jones gas with the following properties: energy flux 5, particle number $N = 5000$, temperature $T^* = 1.4$, density $\rho^* = 0.1$, pressure $p = 0.01$ and a timestep of $t = 0.005s$

$$C_V = \frac{dE \cdot t}{\Delta T} \quad Z = \frac{pV}{NT}$$

For an ideal gas the compressibility factor Z is equal to 1. There is no Boltzmann constant because we use reduced Lennard-Jones units. As seen in our simulation $Z = 0.83 \pm 0.03$. This means our gas can be described as an ideal gas as Z is close to 1.

$$\text{degrees of freedom } f = \frac{2c_V}{R} = 3.03 \pm 202.98$$

We expect the gas to have 3 degrees of freedom because it's a one-atom gas and then we only have the translational degrees of freedom. This fits well to the measurement. The uncertainty is huge due to numerical fluctuations in our simulation as can be seen in the plot. ...

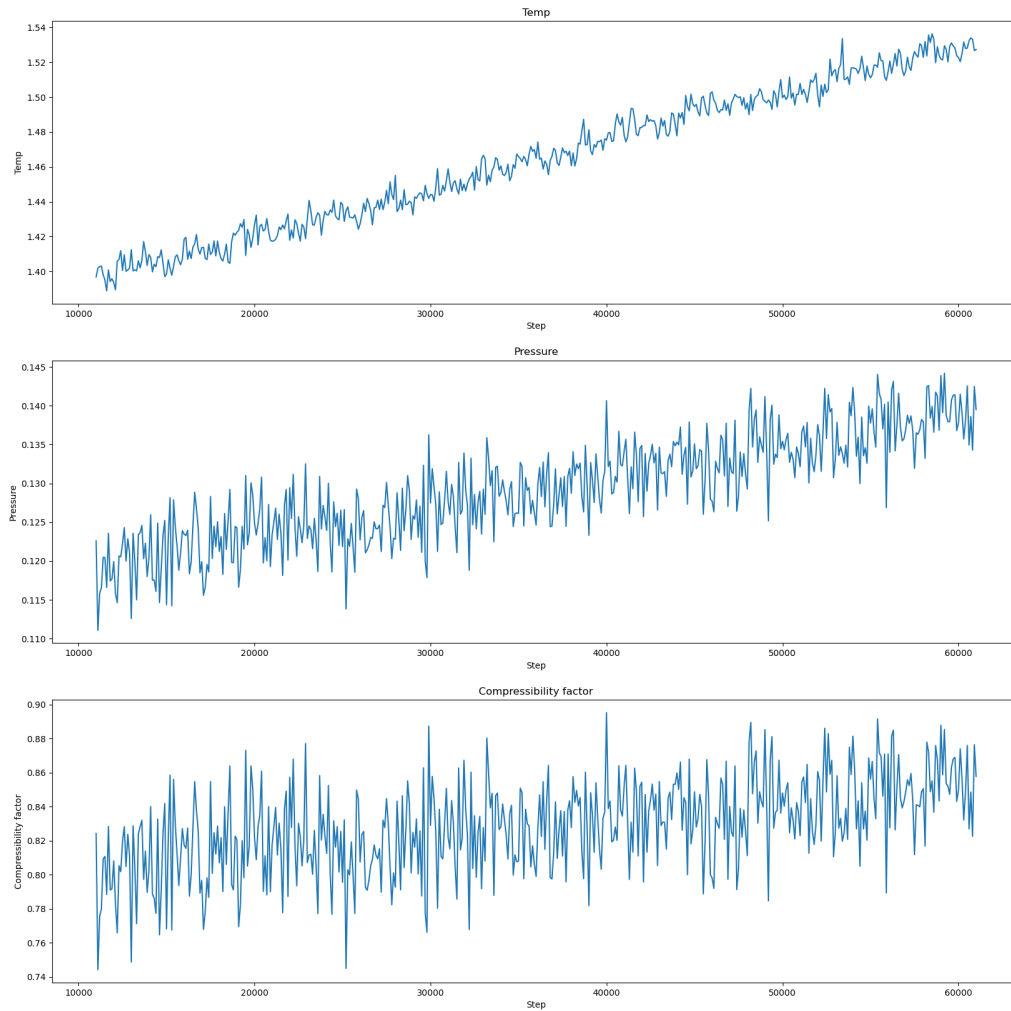


Abbildung 1: simulation

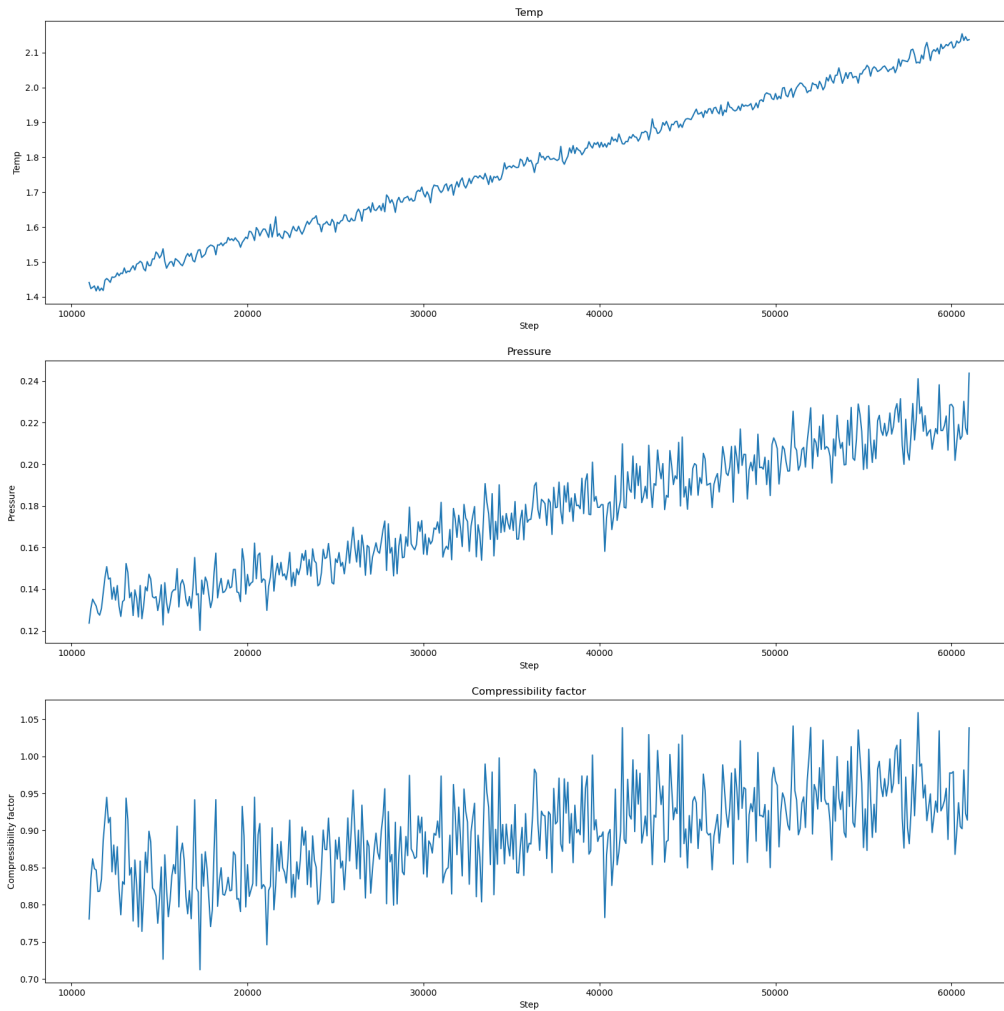


Abbildung 2: simulation with doubled number of particles

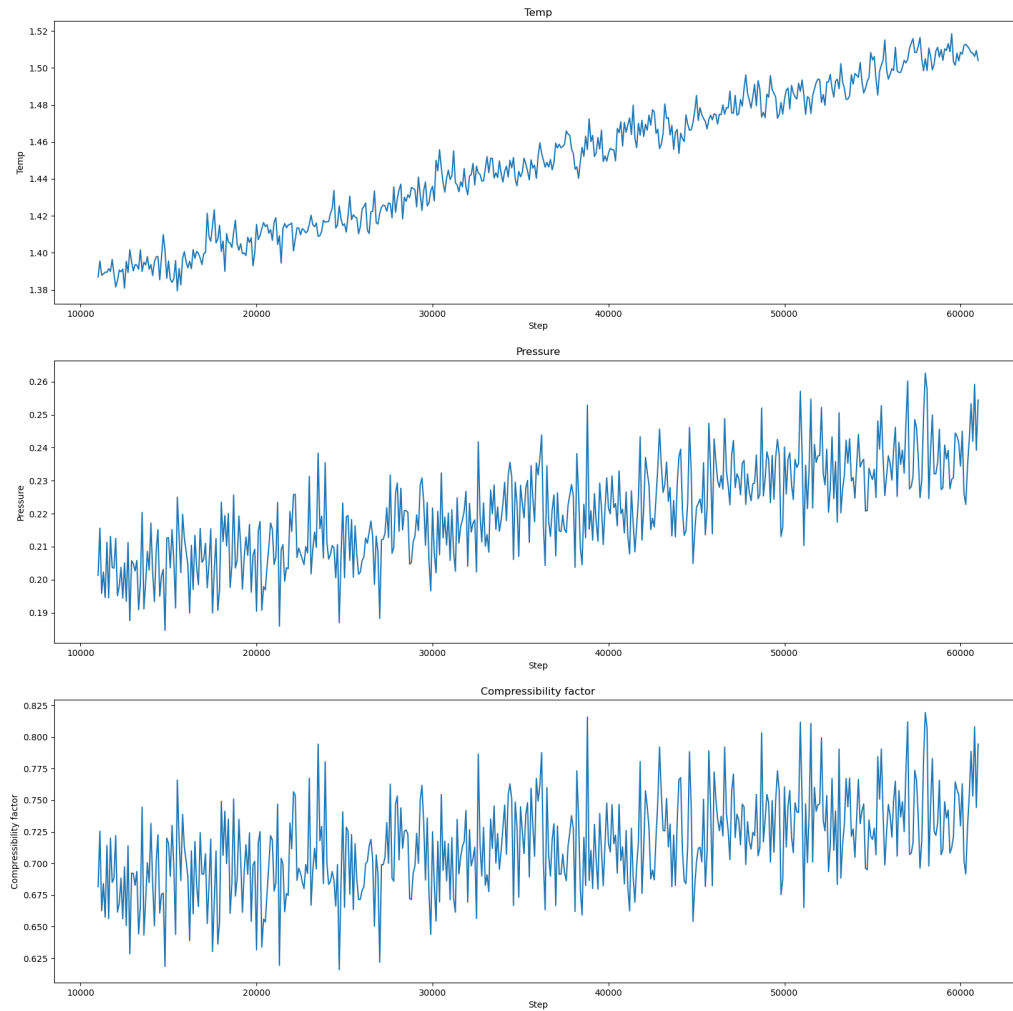


Abbildung 3: simulation with doubled density

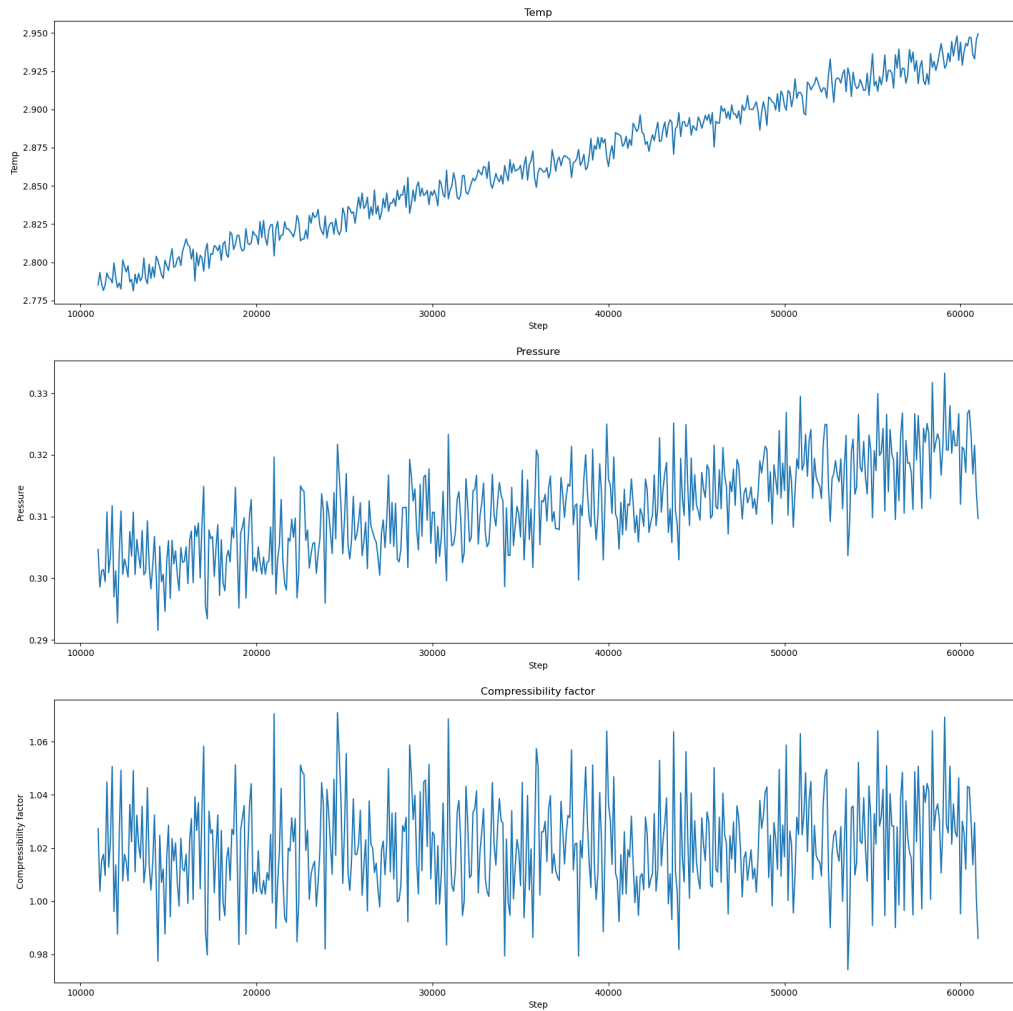


Abbildung 4: simulation with doubled starting temperature

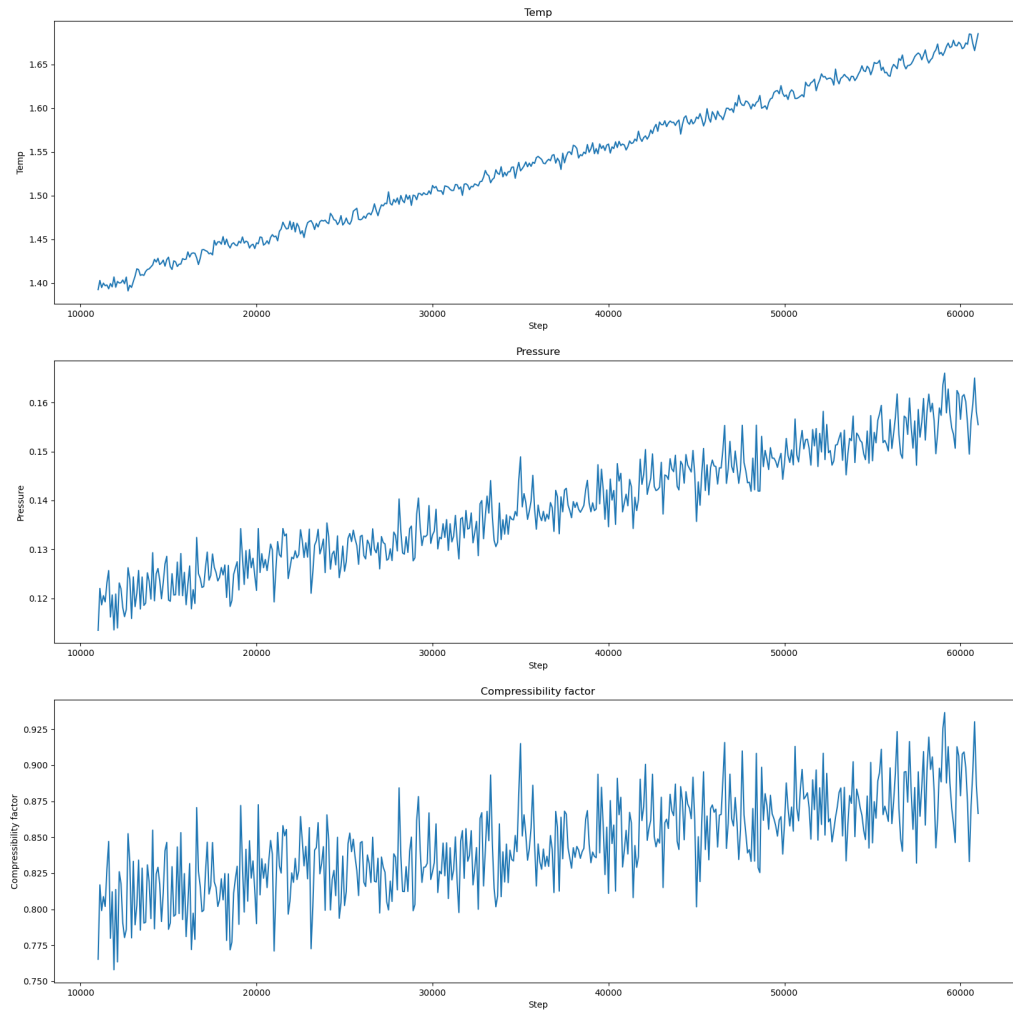


Abbildung 5: simulation with doubled energy flux

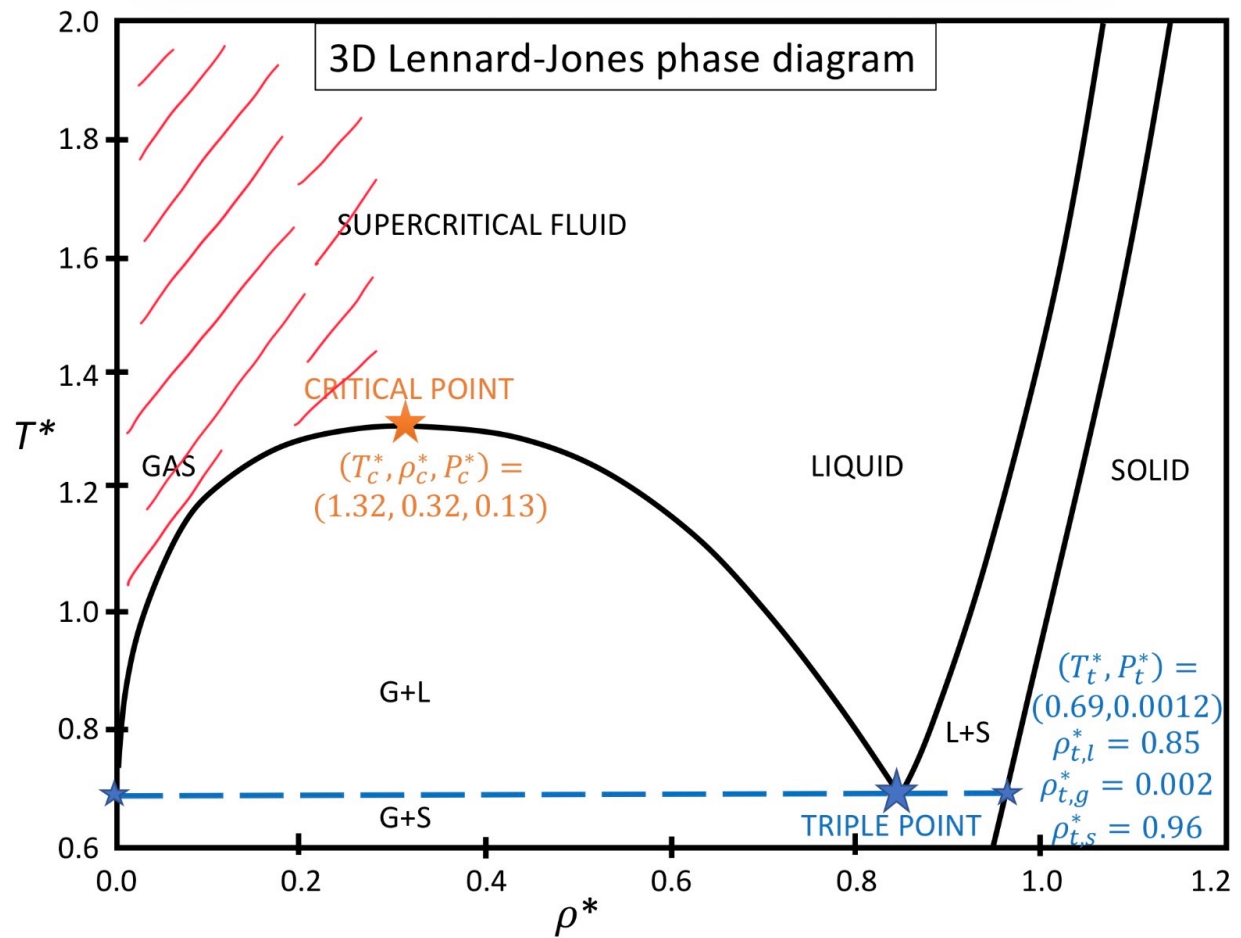


Abbildung 6: Lennard Jones phase diagram: the marked area is the area where our simulations take place

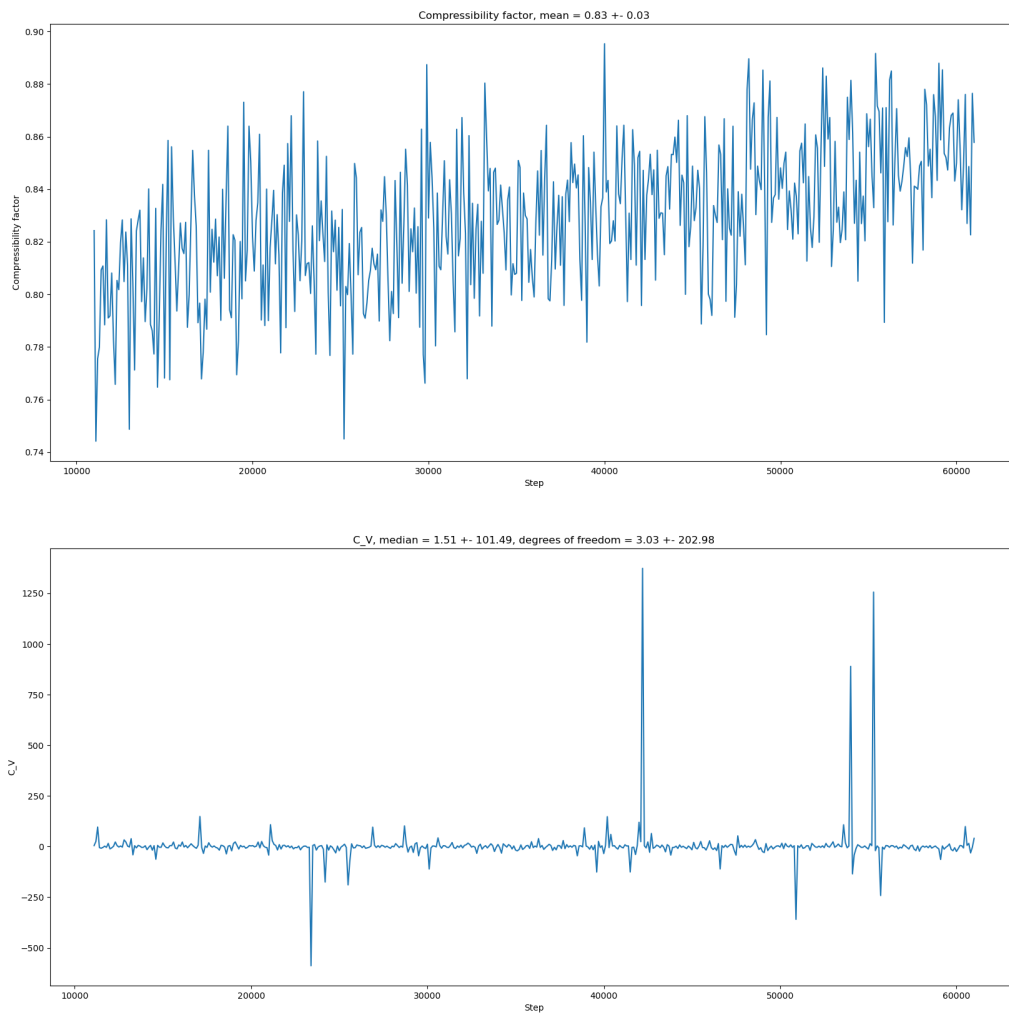


Abbildung 7: Z and C_V

3.3 Different temperatures at fixed gas density

triple point temperature: $T^*_t = 0.69$

critical temperature: $T^*_c = 1.32$

fixed density: $\rho = 0.01$

We chose the following temperatures between the triple point temperature and ten times the critical temperature: $T = T^*_t, 3T^*_c, 7T^*_c, 10T^*_c \in [0.69, 13.2]$

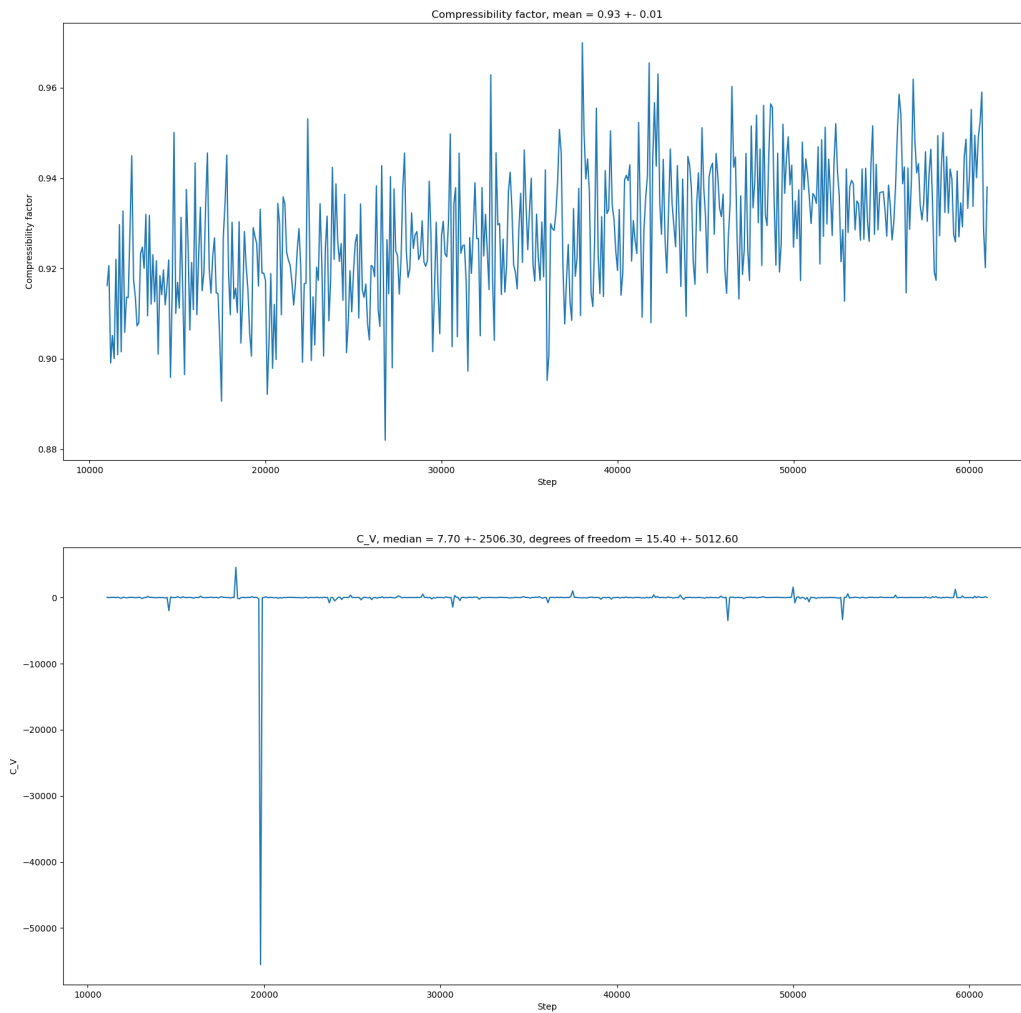


Abbildung 8: Z and C_V for T^*_t

C_V and Z do not change with temperature. For the triple point there was too much fluctuation in our simulated data which is why the median is higher than the heat capacity in the simulations with higher temperatures.

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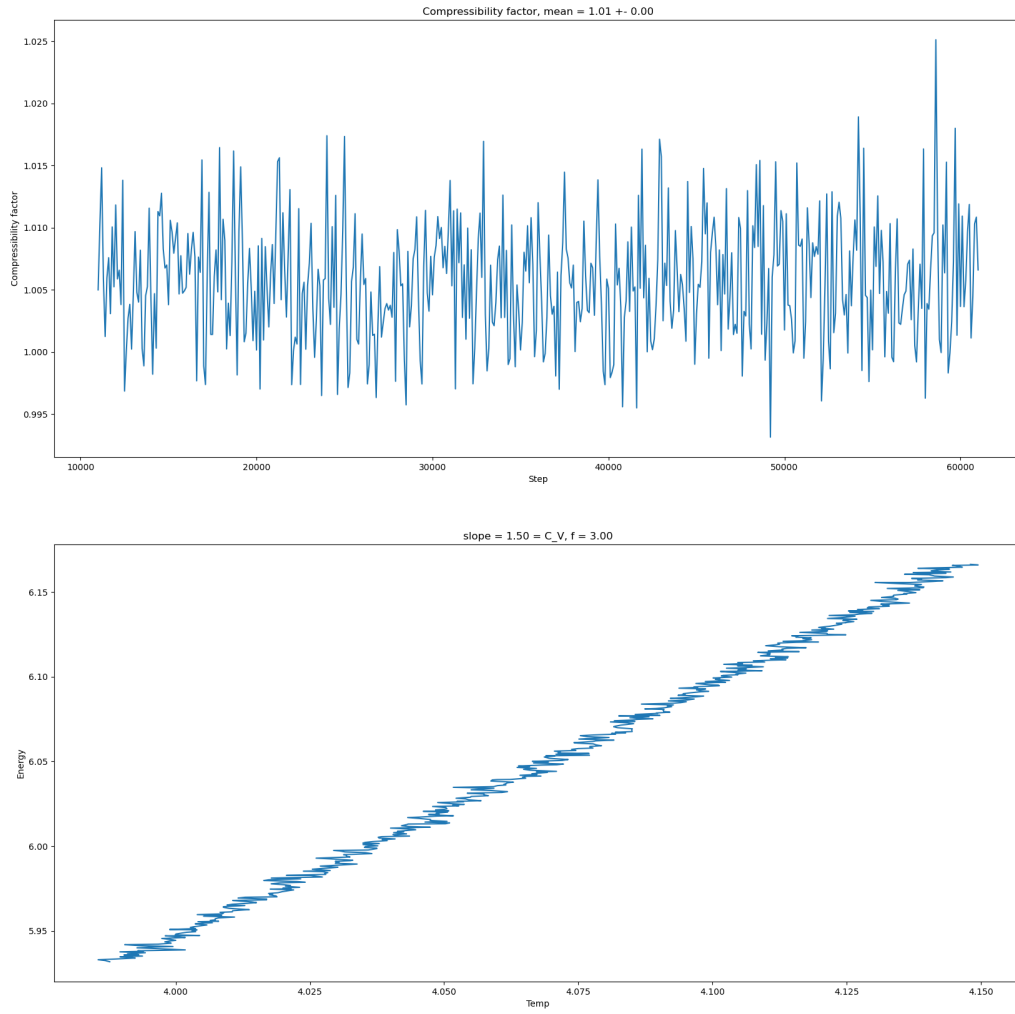


Abbildung 9: Z and C_V for $3T^*_{*c}$

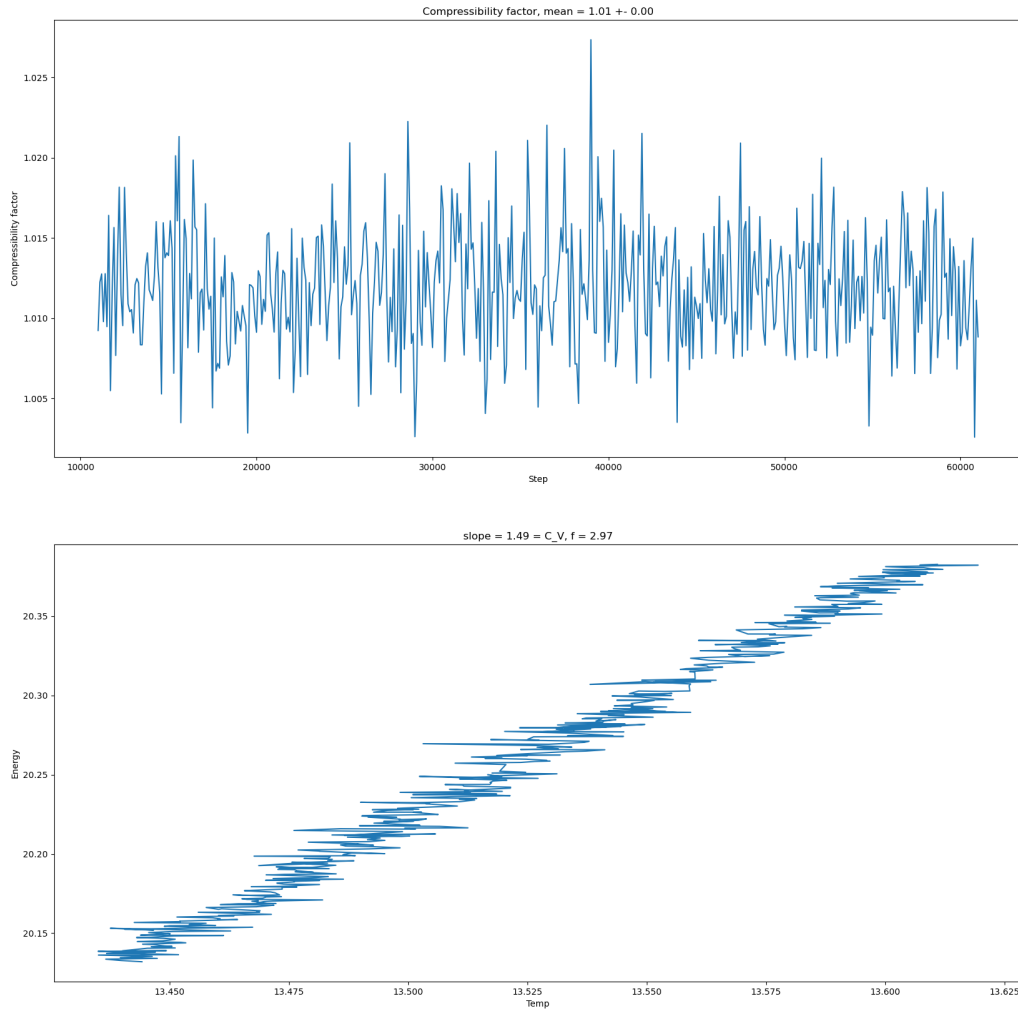


Abbildung 10: Z and C_V for $7T^*_c$

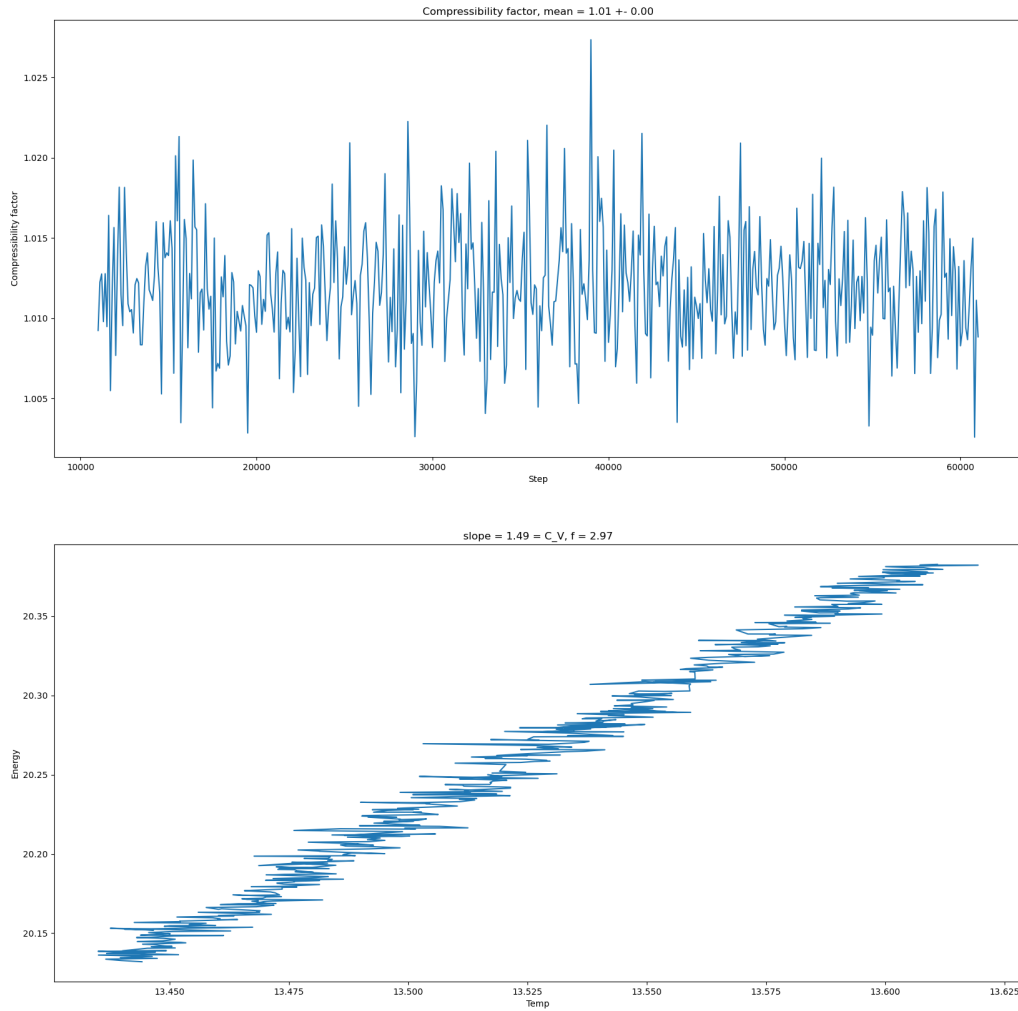


Abbildung 11: Z and C_V for $10T^*_c$

3.4 Temperatures above the critical point

For a temperature $T^* = 2T_c > T_c$ we used the following densities: $\rho = \frac{1}{3}\rho_t, \frac{2}{3}\rho_t, \rho_t$

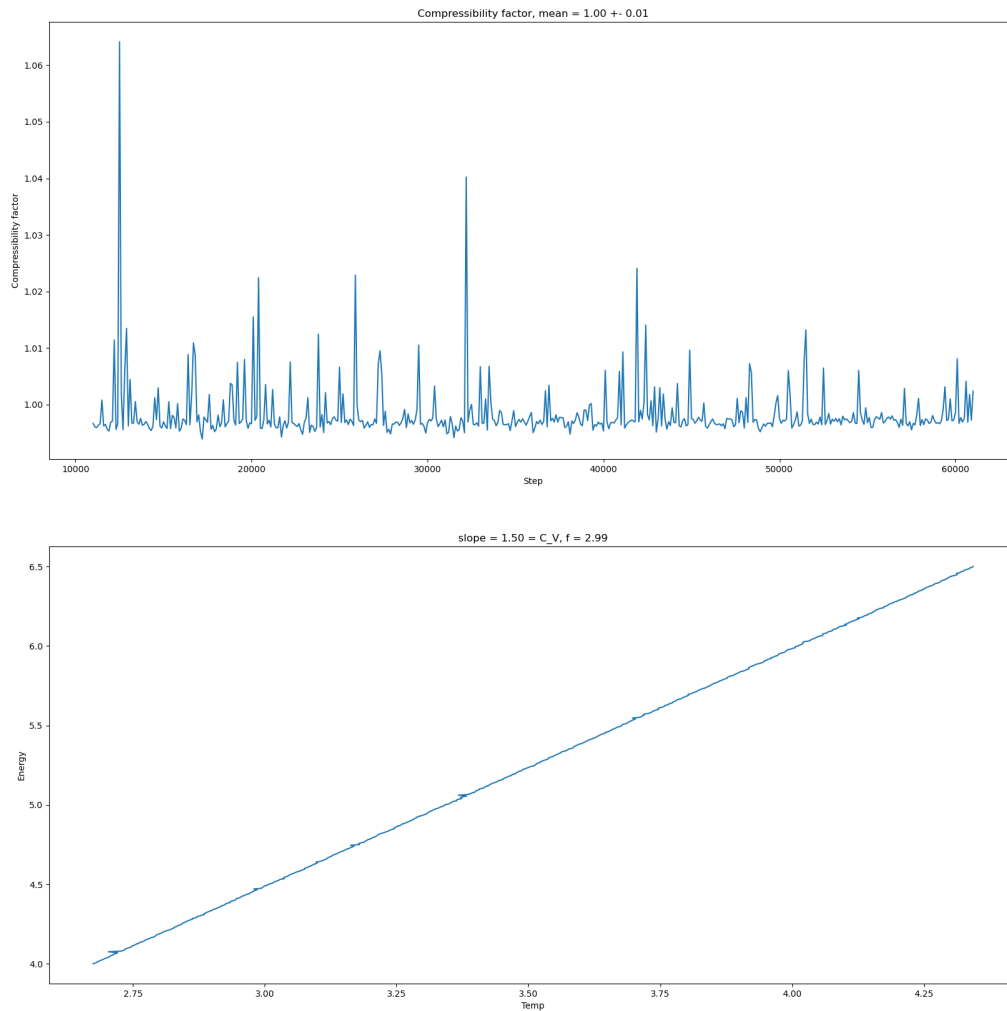


Abbildung 12: Z and C_V for $\frac{1}{3}\rho_t$

For the different densities C_V and Z do not change. ...

3.5 Determination of C_V

The N2 models have 2.56 (rigid) and 3.54 (with vibration) degrees of freedom according to the simulation.

The number of degrees of freedom derived from sound velocity measurements in the lab is $f_{lab} =$

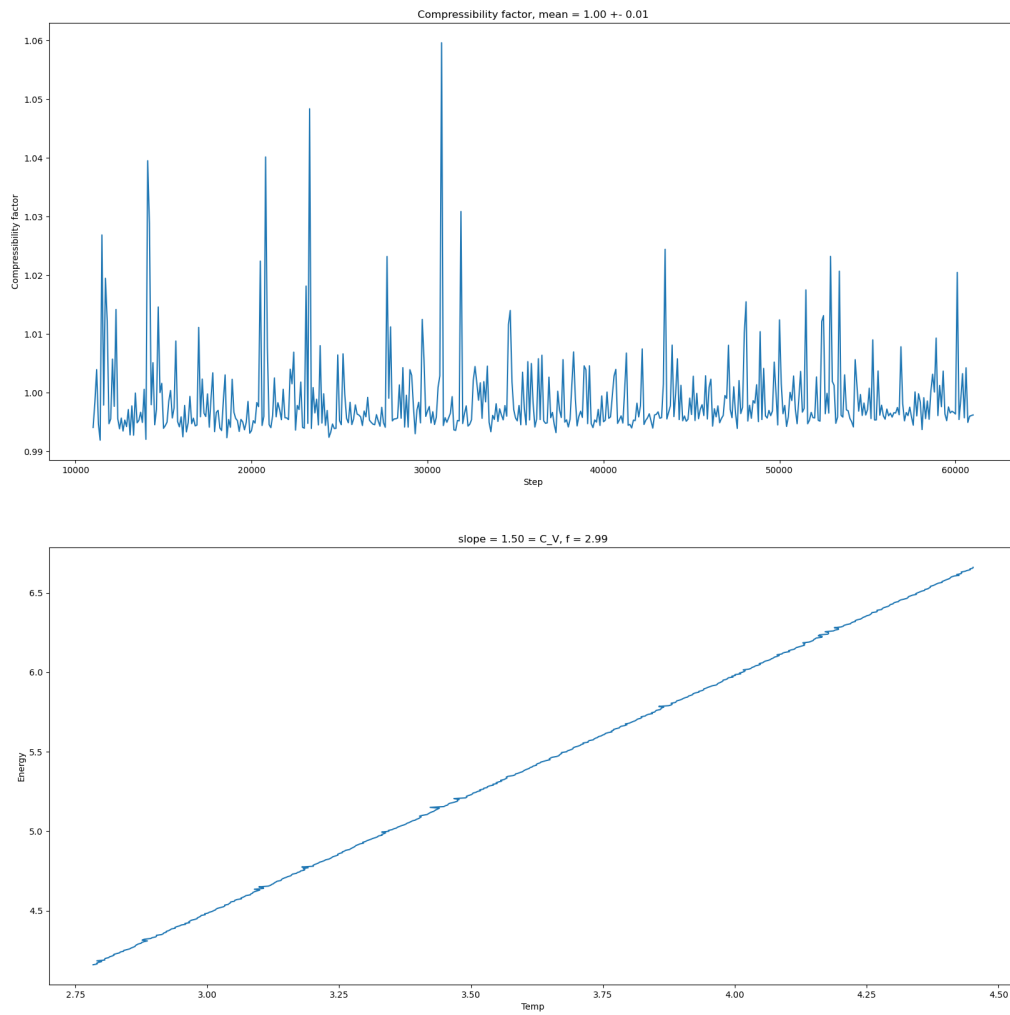


Abbildung 13: Z and C_V for $\frac{2}{3}\rho_t$

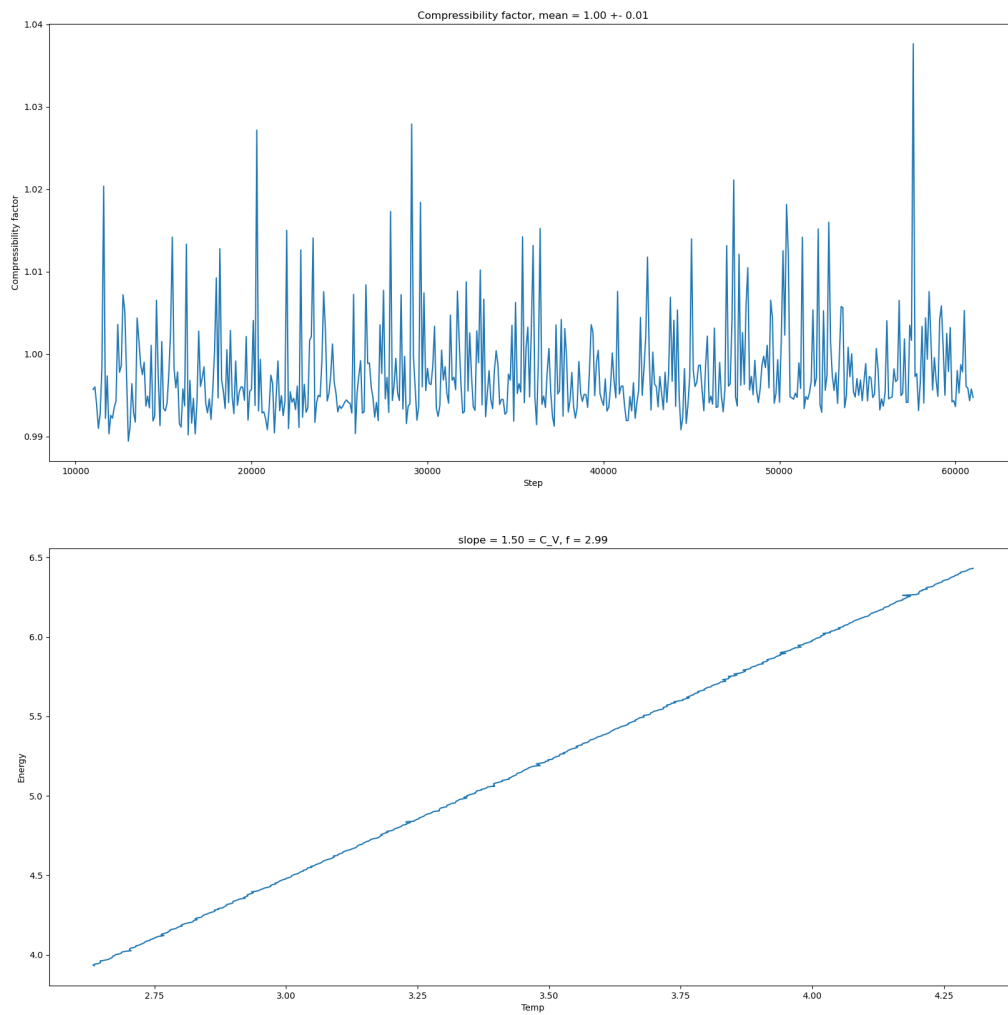


Abbildung 14: Z and C_V for ρ_t

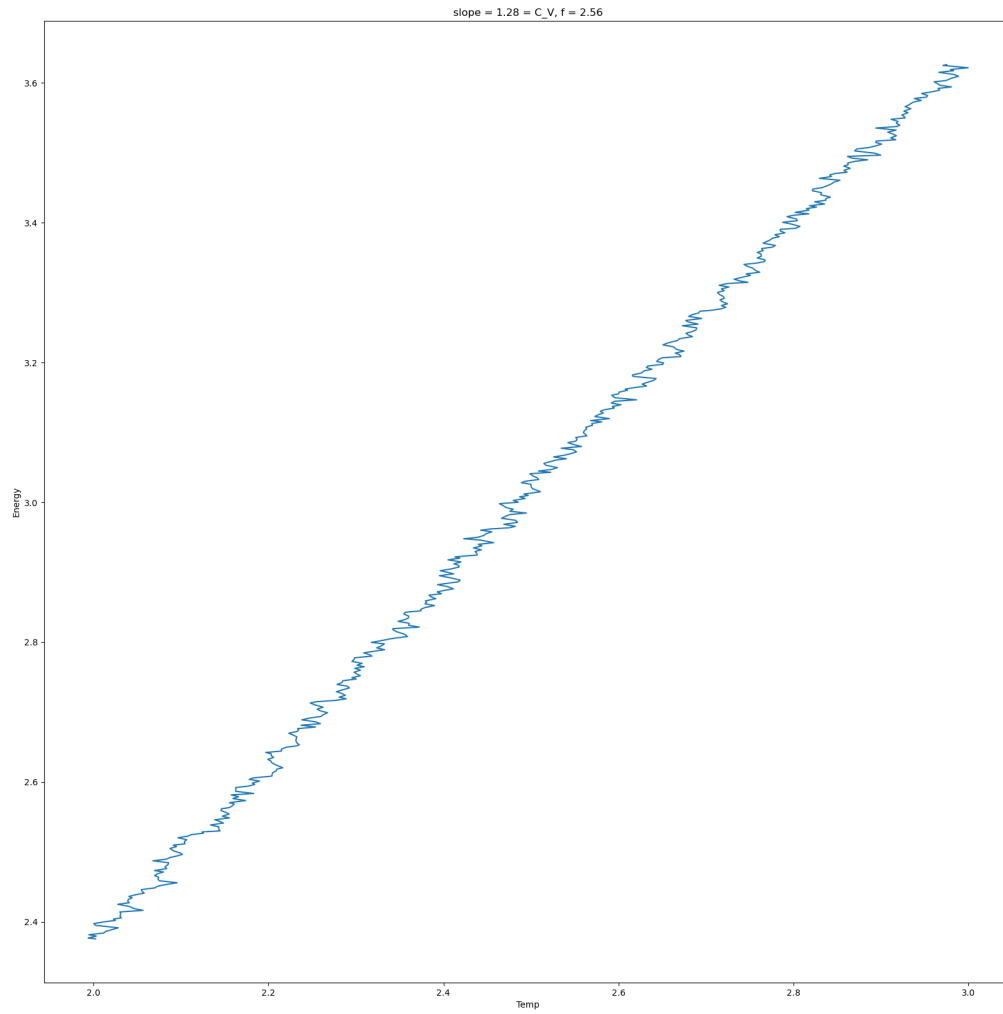


Abbildung 15: degrees of freedom for N2 with rigid bond

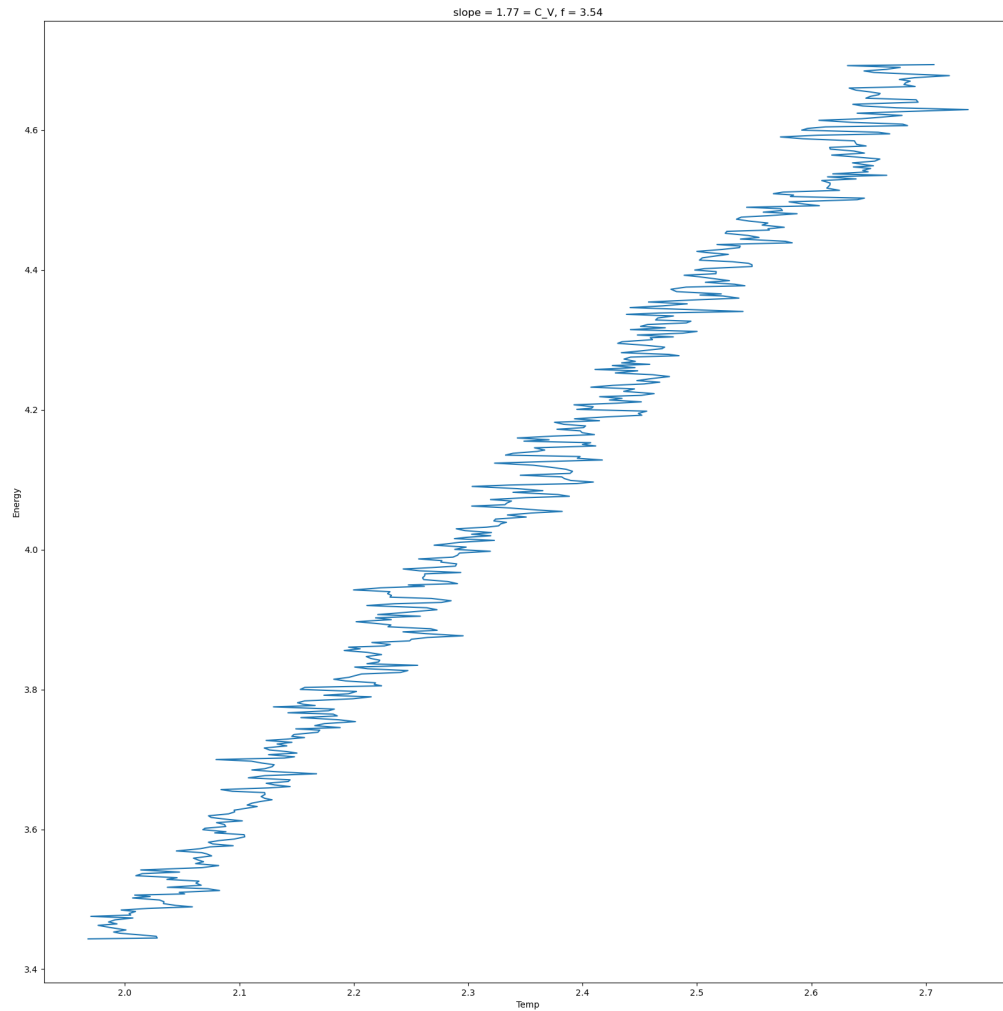


Abbildung 16: degrees of freedom for N2 with vibration

5.81 ± 0.09 , 3.89 ± 0.05 , 4.71 ± 0.07 for air. Since air is made out of 78% N₂ this can be compared to the rigid N₂ which we expect to have 5 degrees of freedom.

We expected f for rigid N₂ to be 5 (3 translation and 2 rotation) and for N₂ with vibrations to be 7 because for the additional degrees of freedom for vibration. This does not fit to our simulations. A reason for this could be that the temperature in our simulation is too low to have the vibrational degrees of freedom.